WEB DEVELOPMENT

Pixel Drift Arcade

A Browser Game Hub for an Indie Studio

Project Brief and Asset Document
Static front end website · HTML, CSS, vanilla JavaScript
Small games. Big drift.

1. Context

Pixel Drift Studio is a small independent team of three developers and one artist who have spent about two years building short, replayable browser games as a side project. The studio now wants one clean public home that introduces the team, displays its catalogue of games in a tidy grid, and lets a first time visitor play a simple game inside the page within seconds of arriving. The studio has a name and a loose retro pixel art mood, but no real website, no fixed colour palette, and no defined typography, so those choices are part of this build.

This milestone is a single static front end website. It is not a game engine, not a backend service, and not a content management system. The site must run entirely from static files served by any plain web server or opened from disk. There is no login, no database, no payment, and no server side code of any kind.

2. Project Information

Field	Details
Project Type	Multi-section static website with one embedded mini-game
Client / Brand	Pixel Drift Studio (fictional indie game studio)
Platform / Medium	Web browser, responsive from 320 pixels wide and up
Deliverable Type	Static site folder plus README.md
Primary Goal	A fast, responsive arcade hub that showcases the catalogue and lets visitors play one game in the page

3. Main Goal

The single most important outcome is a working, responsive, static website that loads the studio brand, lists every game from the provided dataset in a filterable grid, and runs one fully playable keyboard controlled mini-game directly inside the page with a local high score that survives a page refresh. The site must be built only with HTML, CSS, and plain JavaScript, and must open and run without any build step, paid plugin, or internet connection.

1. A branded landing section that introduces Pixel Drift Studio.
2. A filterable catalogue grid built from the provided `games.json` dataset.
3. One embedded, playable canvas mini-game with a saved high score.

4. About Pixel Drift Studio

Pixel Drift Studio is a four person independent team, three developers and one artist, that has been making tiny browser games for roughly two years. No website exists yet, no logo file is provided, no colour palette is fixed, and no typography is chosen. The brand identity beyond the name and the retro arcade mood is open for the freelancer to define within the rules listed in this brief.

Brand tone: playful, retro arcade, energetic, readable, fast to load.
Brand line: `Small games. Big drift.`

5. Requirements

The freelancer will build a single responsive website made of stacked sections on one page, plus one embedded canvas mini-game, using only HTML, CSS, and vanilla JavaScript.

Core Functionality

1. The site is a single HTML page made of stacked sections, navigated by a sticky top bar.
2. The sticky navigation bar links to four sections by name: Home, Games, Play, and Contact.
3. Clicking a navigation link scrolls smoothly to the matching section.
4. The site is responsive and remains usable from a viewport width of 320 pixels up to 1920 pixels.
5. The site runs from static files with no backend, no database, and no build step.
6. Opening `index.html` in a modern browser shows the full site with no console errors.

Home Section

- A hero area shows the studio name Pixel Drift Studio, the brand line Small games. Big drift., and one call to action button labelled Play Now.
- The Play Now button scrolls smoothly to the Play section.
- The hero area includes a short studio summary of two to three sentences drawn from the context above.

Games Catalogue Section

- The catalogue reads its content at runtime from the file `data/games.json` provided in this task.
- Each game renders as a card showing its title, genre, release year, a one line description, and a thumbnail colour block built from the `color` field.
- A filter control lets the visitor filter cards by genre. The genre options are built automatically from the values present in the dataset.

- Selecting a genre shows only cards of that genre. Selecting the All option shows every card.
- The card count visible on screen updates to match the active filter.
- The dataset holds at least ten games and the grid shows all of them when the All filter is active.

Play Section (Embedded Mini-Game)

- The Play section embeds one playable mini-game drawn on an HTML5 canvas element.
- The mini-game is a single avatar that drifts left and right across the bottom of the canvas, controlled by the left and right arrow keys, catching falling blocks that spawn from the top.
- Catching a falling block adds one point to the current score. A missed block that reaches the bottom edge removes one life.
- The player starts with three lives. When lives reach zero the game shows a Game Over message and a Restart button.
- The current score and remaining lives are shown on screen during play at all times.
- The highest score reached is saved to the browser using local storage under the key `pixelDriftHighScore` and is shown on screen.
- The saved high score persists after a full page refresh.
- A Restart button resets the score and lives to the starting values and begins a new round without reloading the page.

Contact Section

- The Contact section shows a static contact block with a studio email placeholder reading `hello@pixeldrift.studio` and a one line invitation to get in touch.
- The Contact section includes a simple form with a name field, an email field, and a message field, plus a Send button.
- The form validates that all three fields are filled before it accepts a submission. An empty field shows an inline message asking for the missing value.
- On a valid submission the form clears its fields and shows an on screen thank you message. The form does not send data anywhere and no backend is involved.

6. Visual and Technical Specs

Dimensions and Layout

- **Canvas game size:** `480 x 360 px` drawing surface, scaled responsively to fit its container.
- **Maximum content width:** `1200 px`, centred on the page.

- **Minimum supported viewport:** `320 px` wide.
- **Card grid:** at least three columns at 1024 px and wider, at least two columns between 640 px and 1023 px, one column below 640 px.

Typography

Usage	Font	Notes
Headings	A geometric or pixel style web safe or open licensed font	Bold weight
Body	A clean sans serif such as Arial or an open licensed equivalent	Minimum 16 pixel base size
Score and game text	A monospace font such as Courier New or an open licensed equivalent	Used inside the game

Not allowed: decorative script fonts, handwriting fonts, and fonts that require a paid licence.

Colour rules

1. The palette uses at most five colours plus black and white.
2. Body text keeps a contrast ratio of at least 4.5 to 1 against its background.
3. The chosen accent colour is reused for buttons, active navigation links, and active filter states.
4. Each game card thumbnail block uses the `color` hex value from that game record in the dataset.

7. Technical Requirements

- **Framework:** none. Plain HTML, CSS, and vanilla JavaScript only.
- **Language:** HTML5, CSS3, and standard JavaScript that runs in current browsers.
- **Platform / Target:** responsive web, working from 320 pixels wide and up.
- **Storage:** browser local storage for the high score only.
- **Data source:** the provided static `data/games.json` file, loaded at runtime.
- **Browser support:** the latest stable versions of Chrome, Edge, Firefox, and Safari.
- **Code quality:** clean, commented, and organised into sensible folders.

Not Allowed

- Front end frameworks or libraries such as React, Vue, Angular, jQuery, or Bootstrap.
- Backend servers, databases, or authentication.
- Paid plugins, paid fonts, or paid assets.
- Build tools that require a compile step before the site runs.

8. Dataset Specification

The catalogue is driven by `data/games.json`, a single JSON array of game records. Each record has the fields below. The seller must not rename these fields.

Field	Type	Meaning
`id`	number	Unique id for the game
`title`	text	Game title shown on the card
`genre`	text	Genre used by the filter control
`year`	number	Release year shown on the card
`description`	text	One line description shown on the card
`color`	text	Hex colour string used for the card thumbnail block

The provided dataset holds at least ten records and at least four distinct genres. All catalogue content on the site comes from this file.

9. Deliverables

#	Item	Format	Notes
1	index.html	.html	Single page holding all four sections
2	Stylesheet	.css	All site styling, in a css folder
3	Site script	.js	Navigation, catalogue, filter, form logic
4	Game script	.js	The canvas mini-game logic
5	games.json	.json	The provided dataset, read at runtime
6	Images or icons used	.png / .svg	Local only, in an assets folder
7	README.md	.md	Setup and run instructions

Folder Structure

```
pixel-drift-arcade/
  index.html
  css/
    styles.css
  js/
    main.js
    game.js
  data/
```

```
    games.json
  assets/
    [local images or icons]
  README.md
```

10. Scope Boundaries

Do

- Build a complete responsive single page site with the four sections described.
- Read all catalogue content from the provided dataset at runtime.
- Build one playable canvas mini-game with score, lives, restart, and a saved high score.
- Use free, open licensed, or self made placeholder assets where images are needed.

Do Not

- Build any backend, database, account system, or payment.
- Add a second page or a routing system. The site is one page with stacked sections.
- Use any front end framework or third party UI library.
- Copy real brand names, logos, characters, or artwork from any existing company or game.

11. Acceptance Checklist

The delivery is complete only if every item passes.

1. The site is one HTML page with four named sections navigated by a sticky bar.
2. All catalogue content is loaded at runtime from data/games.json.
3. The grid shows at least ten game cards under the All filter.
4. The genre filter works and the visible count updates with the active filter.
5. The embedded mini-game is playable with arrow keys and tracks score and lives.
6. Lives start at three and Game Over appears at zero lives with a Restart button.
7. The high score persists after a page refresh through local storage.
8. The contact form validates three fields and clears on a valid submit.
9. The site is responsive from 320 pixels and up with the required column counts.
10. The delivered folder matches the required structure and a README explains how to run it.

12. Final Goal

Success looks like a small, fast, and cheerful arcade website that a first time visitor can open, scroll through to meet Pixel Drift Studio, browse a filterable shelf of the studio games, and within seconds play a simple block catching game whose best score is still waiting after the page reloads. The result feels like a real indie studio front door, built cleanly from static files, easy to host anywhere, and ready to grow as the studio adds more games.